Appendix 15 – Quality Requirement

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1 Abbreviations 3
2 Supplier Quality Responsibilities 3
3 Inspection and Testing 4
3.1 Inspection and Test Plan (ITP) 4
3.2 Inspection Authority 7
3.3 Inspection Notice 8
3.4 Pre-Inspection Meeting (PIM) 8
3.5 Dimension Inspection 9
3.6 Positive Material Identification(PMI) 9
3.7 Welding and Welding Inspection 9
3.8 Non Destructive Examination(NDE) 10
3.9 Pressure test 10
3.10 Pickling and Passivation 10
3.11 Remote Inspection/Meeting 10
4 Coating Inspection & Insulation Inspection 10
5 Mechanical Completion (MC) 11
6 Bolt tightening and Leak Test 11
7 Factory Acceptance Test (FAT) 11
8 Classification Certification 12
9 Approved Vendor List(AVL) 12
10 Punch List and Non-conformance report (NCR) 12
15.1 Punch List 12
15.2 Non-conformance report (NCR) 13
11 Quality Records 13
12 Inspection Release Certificate (IRC) 15
13 Preservation, Packing and Transportation 16
14 Qualification for Operators and Inspectors 17
15 Quality Records Submission 18
16 Quality Assurance 18
17 Material Traceability, Marking and Control. 19

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1 Abbreviations

Abbreviations	Descriptions
CA	Certifying Authority
CPY	Company (PETROBRAS)
FAT	Factory Acceptance Test
ITP	Inspection and Testing Plan
IRC	Inspection Release Certificate
MC	Mechanical Completion
MRB	Manufacturing Record Book
NDE	Non-Destructive Examination
PMI	Positive Material Identification
PIM	Pre-Inspection Meeting
PURCHASER	Hyundai Heavy Industries
QMS	Quality Management System
RFQ	Request for Quotation
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Test

2 Supplier Quality Responsibilities

Supplier shall ensure that all equipment and materials supplied for the Work meet the requirements of the codes, standards, specifications and data sheets, and are sourced from the approved vendor list. If Supplier wishes to deviate from any of these documents, Supplier shall submit in advance a deviation request with justifications. Unless Project agrees otherwise in writing, Supplier shall not deviate from the codes, standards, specifications and data sheets, and approved vendor list.

Supplier shall be responsible to transfer all applicable specifications to Sub-suppliers to comply with project requirement. Drawings and information, i.e. WPS, WPQT, NDE Procedure, Weld Map, NDE Map, Coating Procedure, ITP and Other Test Procedure, etc, from Sub-suppliers shall be submitted for PURCHASER and CPY's review & approval prior to commence of any work by Sub-suppliers.

At a minimum, The following document shall be submitted by Supplier.

- Required drawings including acceptable tolerance.
- Inspection and Test Plan(ITP).
- Data sheets
- WPS, WPQR and approved welder list for welding by supplier and sub-suppliers.
- PMI procedure
- NDE procedure. NDE personal qualification
- Coating procedure. Coating personal qualification
- Other Inspection and Test Procedure
- Dimension Control Procedure
- Bolt Tensioning Procedure

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PURCHASER and CPY's Review, Endorsement and Approval of Supplier's document submission(i.e. drawings, data sheet, procedure, ITP and other documents prepared by supplier/sub-supplier) shall not be construed as a detailed review of the submission and / or as a waiver of any of the technical/project requirements and shall not relieve supplier/sub-supplier of any of its obligations for compliance with all the technical/project requirements.

Supplier shall incorporate the comments and corrective actions raised by PURCHASER and CPY into the relevant documents and procedures. If Supplier disagrees with any of the comments or corrective actions, Supplier shall meet with and resolve such with PURCHASER and Company. Ignorance of PURCHASER and CPY's comments is unacceptable. All comments and corrective actions must be closed prior to the respective work being carried out.

Before attempting to carry out the inspection and especially the checking of calibrations / tests etc. the authorized inspector shall check that the factory test equipment, as supplied by the manufacturer / supplier, is suitable for the testing included in the purchase order. The factory shall provide a current 'Calibration Certificate' with the calibration carried out against primary test equipment. The manufacturer shall be able to prove that equipment used for calibration is traceable back to international standards, with certificates issued by approved independent laboratories, institutes or other bodies.

3 Inspection and Testing

3.1 Inspection and Test Plan (ITP)

Supplier shall provide and implement an Inspection and Test Plan (ITP) for each inspection and testing required in Project. The ITP shall be submitted to PURCHASER and CPY for review and approval. The ITP shall contain inspection involvement to be witnessed by PUCHASER, schedule of inspection and test, test procedures.

Supplier shall ensure that inspection will be carried out for witness and hold activities of PURCHASER and CPY as per approved ITP. At its discretion, PURCHASER may conduct detailed inspections in addition to those mentioned in the approved Test and Inspection Plan.

ITP which supplier submits shall be satisfied with requirement stated in this section as minimum.

Legend of inspection involvement to be used on ITP shall be as follows;

1) "Review–R" : A point at which PURCHASER and/or COMPANY shall review the relevant document as per the approved Inspection & Test Plan (ITP).
2) "Spot Witness-SW" : A point at which PURCHASER and/or COMPANY shall attend or witness the initial activity only. In case any discrepancy is found during initial

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inspection, the number of PURCHASER and CPY's visit will be increased on a random basis. If there is no abnormal issue at the initial activity, further inspection may be waived with declaration of PURCHASER and/or Company then Inspection notification shall be provided continuously.

3) "Witness–W" : A point at which PURCHASER and/or Company shall attend or witness the activity as per approved Inspection & Test Plan (ITP). If there is no notice for participation from PURCHASER and/or Company until appointed inspection date, inspection can be performed without PURCHASER and/or COMPANY.

4) "Hold–H" : A point at which PURCHASER and/or Company shall inspect and complete the activity with participation as per approved Inspection & Test Plan(ITP). Otherwise, no further activity shall proceed without prior permission by PURCHASER and/or Company. PURCHASER and/or COMPANY shall provide notice of attendance intention when inspection notification has been provided.

5) "Monitoring–M" : A point at which PURCHASER and/or Company may visit to observe fabrication progress going on anytime as per approved Inspection & Test Plan(ITP). PURCHASER and/or COMPANY should provide advance notice to supplier for the monitoring on desired schedule.

* Note(1) : When a conflict exists between POS, Company specification and ITP, POS and Company specification have priority. Supplier is not at liberty to assume which instruction shall govern.

Supplier shall develop and implement processes and procedures which efficiently and effectively monitor, verify and document Quality of Work scope deliverables. Supplier shall ensure that supplier and sub-suppliers ITP are prepared at a level of detail sufficient to address all Quality Control related activities in chronological order, from contract review through manufacturing, fabrication, required inspection, assembly, required testing, documentation, and certification. In addition, Supplier shall ensure compliance with the following requirements to be incorporated on ITP as minimum:

- To ensure the compatibility of the project, procedures and applicable internal documents to the equipment or material which constitutes the object of supply;

- To indicate the fixtures and inspection equipment items (including measurement uncertainties) required to achieve quality by checking critical dimensions, functional tests, etc;

- To indicate the personnel qualification performing inspection and checking activities;

- To identify the inspection points foreseen throughout the equipment or material production cycle (for example: raw material delivery, intermediate fabrication steps, painting, intermediate testing, final testing, transport preparation, delivery at site), where checking and inspection shall be performed (mandatory hold points, documents review and witness points), including checking and inspection of sub-suppliers. To indicate the type of examination, trial, test or inspection to be performed.

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- To indicate the execution procedures and acceptability standards for all quality features and requirements, including those, which contain a subjective element. To consider, when so determined in the project, the reference quality level (e.g. API/PSL);

- To identify and develop quality records, mentioning the type of record: report, certificate, graph, etc. Such records shall allow the identification and traceability of critical components along the entire manufacturing process.

- Each Inspection and Test Plan shall comprise, in detail, all the manufacturing stages of the respective equipment or material, as well as their critical components, including those adjudicate by sub-suppliers.

- The aforementioned documents, the documents concerning the treatment of nonconformities and the qualification records regarding welding and welder's procedures shall be available for inspection during the manufacturing process.

- Special attention shall be taken to all equipment / skids / material that are subject to coating application, metallizing, lining and / or material specification, as to ensure that these points are considered in the Inspection Test Plan.

- Supplier shall indicate in the Inspection and Test Plans the extension of its own participation, or by an inspection company which it may have contracted, in the performance of manufacturing inspection, pointing out the document review, the mandatory hold points and the witness points.

- All supplier and sub-suppliers ITP activity shall be performed using a PURCHASER and CPY approved supplier and sub-suppliers ITP.

- Inspection and test activity shall be initiated after PIM & ITP approved.

- PURCHASER and/or COMPANY reserves the right to select witness and hold points within all developed supplier and sub-suppliers QC/ITPs for PURCHASER and/or COMPANY oversight of selected functions and to perform surveillance or audits of the Work.

- When applicable, the following sampling criterion may be adopted by the inspection company:

  STANDARD NBR 5426 Sampling procedures and tables for inspections by attributes.

  - Inspection level II;
  - Simple sampling plan;
  - Normal inspection;

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- Quality limit: 10%;
- Consumer's risk: 5%.

The Supplier shall provide and implement an Inspection and Test Plan (ITP) for each inspection and testing to support the project installation and start-up schedules. The ITP shall be submitted to PURCHASER and CPY for review and approval. Included in the ITP not only covers all significant components and the assembled equipment covers but also the inspections and tests carried out by Sub-suppliers during the manufacture.

Supplier shall ensure that inspection will be carried out for witness and hold activities of PURCHASER and CPY as per approved ITP.

3.2 Inspection Authority

Supplier shall accept the presence of representatives from COMPANY, PURCHASER, other nominated representatives and PURCHASER at Supplier's facilities and at any Sub-suppliers premises, for the purpose of verifying quality, progress check, document review and that Work is in accordance with authority requirements and the requirements in this Contract. PURCHASER reserves the right to participate of any meeting or activities to be held among Suppliers, Classification Society and any other Sub-Suppliers.

PURCHASER and/or COMPANY or its duly authorized representatives shall have access, at all reasonable times, to supplier and sub-suppliers books, records, correspondence, instruction, invoices, plans, drawings, receipts, vouchers and memoranda of every description pertaining to work for the purpose of auditing and verifying the cost of the work or for any other reason associated with work.

At the PURCHASER and CPY's discretion, an in-plant inspection may be performed on the equipment purchased. PURCHASER and CPY shall have free entry, at all times while work on this contract is being performed, to all parts of the Supplier's and Sub-supplier's premises which involve the manufacture, inspection, or testing of the equipment. Supplier shall provide all required inspection tools, instruments, facilities, utility and all up to date engineering / manufacturing documents at free of charge to the Inspector(s). Supplier shall also instruct his Sub-suppliers of this requirement.

Supplier shall get approval the fabrication procedures and PURCHASE shall inspect all materials and equipment supplied by suppliers and sub-suppliers, checking the compliance with Purchase Orders and/or Contract specifications and requirements. Supplier shall make available field inspection teams upon issuance of the Purchase Order and make sure that all contractual specifications applicable to the equipment / system be submitted to the team. Supplier shall issue a report of each inspection performed.

At any time, PURCHASER can reject any materials or equipment supplied by vendors, sub-vendors or subcontractors that do not meet the contract requirements. In case of rejected materials or equipment, Supplier shall replace them, at Supplier's account, within a time period compatible with the time schedule of the contract.

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At any time, PURCHASER may request the substitution of any material and/or equipment not complying with Agreement specifications and requirements, without any additional charge to BUYER.

Supplier shall ensure responsibility for costs associated with PURCHASER and CPY participation in aborted “Hold,” “Witness” point and equivalent work progress visits which are same applied to sub-suppliers. The aborted visit is what caused, from supplier and sub-supplier, by unprepared inspection, rejected inspection and late notice on inspection postponement and cancellation.

PURCHASER and CPY's participation or not in any of these inspections and witnessing shall not relieve Supplier of its obligations. Supplier shall make provisions and facilitate PURCHASER and CPY’s visits.

3.3 Inspection Notice

For all inspections to be witnessed by PURCHASER, the inspection date shall be confirmed to PURCHASER at least twelve (12) Days prior to the date scheduled for each inspection, with respect to vendors located in Brazil, and thirty two (32) Days prior to each inspection for vendors located outside Brazil.

Should any postponement/cancellation become necessary, Supplier shall provide written notice at least 5 working days prior to the originally scheduled date for inspection.

A certified copy of the manufacturer's reports on inspection and testing carried out prior to Company / Contractor’s visit shall be available before or at the latest, at the time of the inspection or verification. Copies of these reports must be retained by the manufacturer for at least five years.

3.4 Pre-Inspection Meeting (PIM)

Pre-manufacturing meetings should be held to clarify the manufacturer's Inspection & Testing Plan / Quality Plan and to define hold and witness points, held before start of production. The PIM should be held with PURCHASER and CPY at supplier and/or main sub-vendor facility where main fabrication shop is.

The PIM shall be held after approval of the following documents as minimum Code 2(approved with comment).

- WPS/PQR requested
- Approved Welder List requested
- GAD requested
- Data Sheet requested
- ITP

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3.5 Dimension Inspection

Supplier shall put in place a procedure on dimensional control and strictly adhere to it. Method for verification of dimension shall be described in the procedure clearly and the procedure shall be submitted for PURCHASER and CPY's review and approval. Fabrication tolerance approved by PURCHASER and CPY, which has to be shown supplier's drawing, shall be checked at every stage and at the final survey. Especially, Dimension(X, Y and Z coordination) with agreed tolerance for interface points connected to PURCHASER such as skid foundation, piping tie-in and E&I tie in shall be described on approved drawings and carefully measured. The final survey shall fall within the tolerance allowed. PURCHASER and CPY shall be engaged for intermediate and final dimensional survey. These inspection stage shall be reflected on Supplier ITP with PURCHASER witness point.

Supplier shall submit dimension records inspected right after finishing the inspection. Refer to Attachment #5 for an report format of piping tie-in.

In case any non-conformity issue out of tolerance is found against the approved drawings, Supplier shall take corrective actions before shipment.

For accurate measurement, Supplier shall take into account to utilize Digital Dimension Measuring Equipment, e.g. Total Station.

3.6 Positive Material Identification(PMI)

Supplier shall follow the project requirement of PMI based on I-ET-3010.00-1000-970-P4X-002 (REQUIREMENTS FOR NDT) and the all referenced project specifications in the aforementioned specifications.

3.7 Welding and Welding Inspection

Supplier shall follow the project requirement of welding in the package based on I-ET-3010.00-1200-955-P4X-001 (WELDING) and I-ET-3010.00-1200-955-P4X-002 (REQUIREMENTS FOR WELDING INSPECTION) and the all referenced project specifications in the aforementioned specifications.

Welding inspector qualification shall meet requirement in I-ET-3010.00-000-970-P4X-001 (REQUIREMENTS FOR PROCEDURES AND PERSONNEL QUALIFICATION AND CERTIFICATION) and the all referenced project specifications in the aforementioned specifications.

All Welding shall be visually inspected by PURCHASER.

Each welding joint shall be traced on actual product in comparison to welding traceability sheet including the following information

- WPS No. / Welding Date / Joint No. / Welder No. / NDE date / NDE Result / NDE

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Type

3.8 Non Destructive Examination(NDE)

All NDE Procedure, NDE operator and inspector qualification what will be used for this contract shall be submitted to PURCHASER and CPY for approval prior to NDE.

All NDE Reports shall be reviewed by PURCHASER and qualified NDE personnel engaged by supplier and as for RT films, all RT films shall be interpreted by PURCHASER and qualified NDE personnel engaged by supplier at suppler NDE shop or wherever have NDE interpretation facility.

3.9 Pressure test

All piping systems shall be subjected to pressure test except atmospheric vents and atmospheric drain systems. Test procedure shall be approved by PURCHASER and CPY. Vessels and tanks which require hydro test are the same.

Before commencement of pressure test, required welding visual inspection and quality records review such as raw material certificate, PMI report, welding records, NDE reports including RT films and calibration certificate for instrument shall be completed by PURCHASER.

3.10 Pickling and Passivation

For stainless steel and duplex steel piping, pickling & passivation shall be done.

3.11 Remote Inspection/Meeting

Supplier shall provide/devise some sort of method/software/tool/technology which should be used for inspection/meeting in case of the access to facility is restricted to external personnel due to COVID-19 emergency lockdown.

4 Coating Inspection & Insulation Inspection

BIDDER shall follow the project requirement of painting for the package based on I-ET-3010.00-1200-956-P4X-002 (GENERAL PAINTING), I-ET-3010.00-1200-956-P4X-001 (QUALIFICATION TESTS FOR PAINT SYSTEMS) and the all referenced project specifications in the aforementioned specifications

Before commencement of any painting work, Coating Procedure Test(CPT) shall be firstly completed including adhesion test(Min 5MPa) for each painting system applied for the equipment and materials.

BIDDER shall follow the project requirement of qualification of personnel carrying out painting work and inspection based on I-ET-3010.00-000-970-P4X-001 (REQUIREMENTS

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FOR PROCEDURES AND PERSONNEL QUALIFICATION AND CERTIFICATION) and the all referenced project specifications in the aforementioned specifications.

Painting Shop Audit by PURCHASER's coating inspector shall be carried out before starting Painting work including sub-supplier's painting shop.

Where insulation is required, equipment, vessels and piping that require insulation shall be insulated in accordance with I-ET-3010.00-1200-431-P4X-001 (THERMAL INSULATION FOR MARITIME INSTALLATIONS). Insulation shall be suitably protected to prevent transport related damage.

5 Mechanical Completion (MC)

Upon completion of erection, Mechanical Completion(MC) shall be carried out prior to Factory Acceptance Test (FAT). Piping system installation, including valves, and E&I installation shall be walkdown using latest version of approved drawing such as P&ID, GA drawing, Detail drawing, Wiring diagram etc. VENDOR's standard mechanical completion check sheets are to be detailed for the overall package, as well each individual component (e.g. Generator set, Motor, Inter stage & final stage cooler, Pre-filter & After-filter, LO system, Control panel(s) etc).

Before visit of PURCHASER & COMPANY for MC, VENDOR has obligation to provide readiness for inspection. VENDOR shall provide internal MC records in advance.

Inspection on inside of vessels/tanks which are installed on skid shall be carried out with PURCHASER at this stage then an report shall make available with photos.

Punch items which are found during MC shall be compiled/recorded with relevant photos using PURCHASER's format.

6 Bolt tightening and Leak Test

Bolt tightening and Leak Test shall be verified with PURCHASER through torque tightening inspection using torque tool on flanged joints. A report on Bolt tightening for each joint and Leak Test shall be made available and signed off by PURCHASER and submitted.

7 Factory Acceptance Test (FAT)

FAT procedure (including performance test) shall be developed by vendors or sub-vendors and submitted to PURCHASER for approval, at least three (3) months prior to a FAT planned. FAT shall be a part of the vendors or sub-vendors scope of supply and shall be witnessed by PURCHASER and Classification Society surveyor, as applicable.

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Before performing FAT, punch lists which are found during MC shall be closed with PURCHASER witness.

Pre-FAT by supplier for readiness provision to PURCHASER shall be carried out internally.

All test required as a part of FAT shall be tested as a part of Pre-FAT tests to ensure that the entire system complies with project requirements and is ready for FAT. All findings shall be recorded in a punch list and shall be rectified prior to FAT. The entire document including punch list clearance shall be submitted PURCHASER when submitting FAT inspection notification. Pre-FAT shall be observed by PURCHASER if required.

All inspection/test reports for the FAT shall be endorsed/signed by all relevant parties.

Punch items which are found during FAT shall be compiled/recorded with relevant photos using PURCHASER's format. All punch lists generated from the FAT shall be inspected under witness of PURCHASER and signed off for close out.

8 Classification Certification

Classification and/or Certification requirements shall be included in all contracts between PURCHASER and the vendors and/or the sub-vendors. Supplier shall issue an authorization to Classification Society allowing direct and full access of PURCHASER and CPY to all information regarding certification of project and fabrication of equipment and materials. Supplier shall make sure to provide the classification certificate on equipment and materials.

9 Approved Vendor List(AVL)

Equipment/material shall be procured as per Approved Vendor List(AVL) provided by Project. Supplier shall assure the equipment/material commonality and shall obtain Project's approval if anytime it intends to deviate from this list/purchase from other vendors not in the list.

10 Punch List and Non-conformance report (NCR)

15.1 Punch List

Punch list found during time of inspection as per approved ITP shall be filled out using PURCHASER format along with corresponding photos.

Basically, all punch lists shall be closed prior to IRC issuance. If some punch items need to be carried over to PURCHASER's yard due to impossibility of close-out at Supplier shop, Supplier shall get approval for each punch item from PURCHASER. Punch handed over to

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PURCHASER's yard does not relieve any responsibility of Supplier to close it out.

15.2 Non-conformance report (NCR)

Non-conformance Report(NCR) for repair, acceptance or re-grade resolution strategies, together with relevant procedures, must be formally submitted to and accepted by PURCHASER and CPY.

NCR shall include root cause, corrective action and preventive action for the non-conformance issue. All repair work for NCR shall be started after getting approval of Supplier's NCR with repair procedure. NCR shall be formally closed after inspection by PURCHASER and CPY for repaired non-conformance issue.

NCR registration shall be maintained as live document and submit to PURCHASER when requested.

11 Quality Records

All quality records shall be promptly reviewed right after completion of each inspection stage.

Original documentation shall be made available for all tests carried out on the equipment or materials supplied.

Test Certificates and performance records shall be completed using the Supplier and Sub-supplier standard forms. Type of test and equipment/material tag number shall be clearly shown on each document.

Equipment or material suppliers shall keep available for PUCHASER's inspection and auditing at least the quality records indicated below, reinforcing or complementing those required by the technical documents, which shall be inserted in the Data Book accompanying equipment or material items..

- Certificates of metallic and non-metallic materials, including mechanical, physical and chemical tests of raw materials and components to be defined in the analysis of the Quality Plan;
- Records of heat treatment for stress relief, including graphics;
- Records of heat treatment, including dehydrogenation for coated screws and nuts in case of tensile strength in excess of 100 Ksi;
- Hardness test record;
- Ready made inspection record;
- Non-conformities records or reports brought up in the process of manufacturing,

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assembling, tests and expedition, including respective manufacturer's judgment aimed at corrective action.

- Certificates of gauging for all instruments (mechanical, electric, electronic, etc.) used for inspection and tests along the manufacturing process, as well as for the final factory tests.
- Repair procedures and repair records (defects, repaired during manufacture), when applicable;
- Reports concerning visual and dimensional tests;
- Qualification records of welding procedures, welders and welding operators, issued by inspectors qualified and certified by a nationally acknowledged entity in the country where manufacturing occurs;
- Welding reports, including instructions for welding execution and inspection;
- Certificate of welding consumables;
- Records of personnel qualification for non-destructive tests,
- Qualification records of non-destructive test procedures, performed by a qualified inspector;
- Non-destructive and destructive tests reports;
- Hydrostatic and pneumatic tests records;
- Gauging records for peripheral instruments;
- Static and dynamic balancing records of the of the revolving set for rotating equipment;
- Mechanical running tests, including noise level test for rotating or dynamic equipment;
- Performance test records, including vibration measurements for rotating or dynamic equipment;
- Assembly clearance records, as well as records concerning any non-conformity found during disassembling after the tests for rotating or dynamic equipment;
- Electrical/electronic equipment conformity certificates for protection types applied in potentially explosive atmosphere and protection grades;
- Functional tests of electric panels, such as: switchboard, control and protection boards, and their respective components;
- Routine tests certificates for electrical equipment and special test certificates and

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for the respective prototypes;

- Gauging and functional tests records for instruments and automation systems;
- Painting scheme applied and record of painting application quality control
- Preservation, packing, transportation and storage procedures;
- Record of the follow-up inspections concerning painting, detailing surface preparation, each coating applied, as well as the results regarding thickness and adherence; and
- Installation, operation and maintenance manuals.

12 Inspection Release Certificate (IRC)

PURCHASER shall inspect equipment / material at the vendors or sub-vendors site/factory and issue the Inspection Release Certificate (IRC), before shipment, complying with all Purchase Order requirements. In case of any punch lists, vendors and sub-vendors shall present to PURCHASER's approval an action plan with a deadline to clear the issues.

This Punch List and its action plan shall be attached to the IRC and incorporated at Integrated Punch List.

Supplier shall issue a photographic report, demonstrating all items of the packing list, before packing and transportation.

Supplier shall inform PURCHASER when the equipment/skid/package is ready for transportation. PURCHASER at its discretion will attend the release inspection before transportation. The advanced notice for the inspection herein shall be forwarded to PURCHASER thirty (30) days before the event. Additionally, The IRC shall also be part of project records.

Supplier shall ensure, before packing for shipment IRC submission, that equipment / skids / material is painted according to PETROBRAS Painting Technical Specification (I-ET-3010.00-1200-956-P4X-002) or other painting scheme accepted by Petrobras, so they can withstand with weather and marine environment (Unit operation conditions).

All Quality records shall be reviewed before issuance of IRC. Unless there are quality records not reviewed by PURCHASER, IRC will not be issued. If there is additional visit required due to not reviewed quality records after equipment release, the visit cost will be charged to Supplier.

Even though IRC issued amicably, Supplier's responsibility is not relieved for findings or issues at PURCHASER yard and offshore which is not complied to this Contract and Specification.

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13 Preservation, Packing and Transportation

Supplier shall submit packing and preservation procedure prior to shipment for PURCHASER and CPY's review and approval

All packing and preservation in ITP shall be a witness point for PURCHASER inspection – review of Suppliers report may not be acceptable. Then, packing list shall be signed by PURCHASER compared to actual items including spare parts during inspection to check quantity and final appearance status based on packing list. Supplier's check list and/or inspection report including pictures for the packing & preservation shall be developed and provided.

Supplier shall ensure that all equipment, materials, special tools and spare parts that Supplier purchases hereunder are suitably packed, crated, preserved, boxed or otherwise appropriately prepared for shipment and protected from the elements.

Preservation and packing shall be proper for transportation and storage in a marine environment and protected against moisture and damage during transport, handling and lifting, in accordance with vendors and sub-vendors recommendations and specification. In any case, suitable preservation and protective measures shall be provided to prevent equipment deterioration prior to entering into service in the UNIT. The fumigation of the boxes shall be considered, where applicable.

All packing shall be clearly marked for shipping, including lifting points, weight, dimensions and center of gravity. All sea fastening and temporary supports used on the equipment for shipment shall be clearly identified.

Supplier shall ensure that all loose valves, tubes and instruments are supplied with plastic caps. Supplier shall also ensure that all electric panels and motors will be supplied with Volatile Corrosion Inhibitor (VCI) impregnated plastic protection or similar, and external plug for space heater connection.

Supplier shall provide clear and comprehensive instructions on the exterior of all packages advising the necessary warning notices for unpacking, handling and installing the equipment on arrival at destination.

Supplier shall obtain the IRC from PURCHASER prior to shipment. Shipment cannot be proceeded without IRC, any approved punch and approved NCR. The IRC will be issued after PURCHASER's satisfactory for final inspection including punch list, NCR and packing & preservation. In case any abnormal issue is found, Supplier shall take corrective actions before shipment.

Supplier shall submit the following shipping documents (Dispatch Dossier), containing at least:

a) Inspection Release Certificate(IRC), issued by SELLER or its representatives, which

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releases the equipment/material shipment;

b) Certificate of the Classification Society, if applicable;

c) Conformity Certificate;

d) Emergency card or "Material Safety Data Sheet", if applicable;

e) Requirements and instructions for lashing, protection, transportation, unloading and storage, whenever necessary;

f) General Arrangement and Data Sheet;

g) P&ID;

h) Packing List and Invoice

i) Lifting Scheme, if applicable

j) Packing, unpacking, preservation and storage procedures. Such documentation must cover all phases of assembly and testing on Unit.

k) Painting/coating repair procedures.

l) Fumigation Certificate, if executed;

14 Qualification for Operators and Inspectors

Supplier shall follow the project requirement of qualification of personnel carrying out fabrication, construction, assembly and inspection of all equipment, piping, pipeline and structures within the contractual scope of work including sub-supplier based on I-ET-3010.00-000-970-P4X-001 (REQUIREMENTS FOR PROCEDURES AND PERSONNEL QUALIFICATION AND CERTIFICATION) and the all referenced project specifications in the aforementioned specifications.

The final inspection at vendor’s facility shall be carried out with Supplier’s technicians Brazilian Regulatory Standard NR-13 and I-ET-3010.1M-1200-500-PAH-001 Compliance with NR-13 Requirements for Pressure Vessels.

Inspections and hydrostatic test of all pressure vessels and equipment under Brazilian NR-13 shall have documented evidence by means of a report signed by Supplier’s NR-13 Qualified Professional ("Profissional Habilitado") registered with the Brazilian Engineering Council, as per item 13.3.2 of NR-13. Tests shall be witnessed by Supplier´s Qualified Professional or a professional delegated by the Qualified Professional.

Supplier shall support PURCHASER inspection expert during the execution of the final internal and external inspection (Initial Safety Inspection as mentioned in NR-13) at their final location on the equipment.

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15 Quality Records Submission

Quality Records which are compiled to Manufacturing Record Book (MRB) are as follows, but not limited to;

- Non-conformance reports and punch lists
- Manufacturer's mill certificates
- Mechanical Test certificates
- Records of dimensional control reports and other tests
- Records of witnessed tests
- PMI Record
- NDE Test Record
- Material and Equipment certificates
- Welding qualification certificates including welding parameter check sheet, welding traceability sheet and approved welder list
- Welding map, NDE map.
- Weld examination reports
- Protective coatings report
- Details of any repairs
- Heat treat records and charts
- Pressure test reports
- Mechanical Completion Reports.
- Alignment test reports for motors and pumps
- NDE personal qualification certificate
- Painting operator/inspector qualification certificate
- Flange Management System Records

16 Quality Assurance

The Supplier shall demonstrate that they operate a quality management system in accordance with the ISO 9001: 2015 requirements. The effectiveness of the quality system and the Supplier's compliance with it shall be subject to monitoring by PURCHASER and in addition, may be audited following an agreed period of notice.

The quality management system shall provide for the planned and systematic control of all quality related activities performed during design / development, construction, procurement, installation or servicing (as appropriate to the given system). Implementation of the system shall be in accordance with Quality Manual and Project Specific Quality Plan, which shall together with all related / referenced procedures, be submitted to PURCHASER and CPY for review, comment and approval as required by the Purchase Order. The Supplier shall provide facilities for and cooperate with PURCHASER and during manufacturing, assembly, testing and audit.

PURCHASER and CPY reserve the right to conduct a Supplier Quality Audit to verify. Supplier QMS complies with relevant quality standards.

Should Supplier QMS be found deficient during audits or verification visits, Supplier may be given the opportunity to carry out corrective action within a defined period of time. PURCHAER and CPY reserve the right to conduct additional audit visits to verify Supplier

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compliance with corrective actions.

Additional process and deliverables quality verification visits may be employed by PURCHASER and COMPANY to monitor Supplier Quality Control.

Supplier shall provide 12 (Vendors in Brazil) or 32 working days (Vendors outside Brazil) to PURCHASER regarding actual date and location for all the audits to be undertaken to permit PURCHASER and CPY. If PURCHASER and CPY decides to re-audit any portion of the Supplier's quality system, Supplier shall cooperate all necessary assistance and access

All quality records such as Material Certificate, PMI report, welding record, NDE record, pressure test report, coating report, MC check sheet, FAT report, packing & preservation report and packing list shall be documented by Supplier and reviewed by PURCHASER

17 Material Traceability, Marking and Control.

All materials and equipment supplied by Supplier shall be brand new (not overhauled), unused, fully traceable, field proven, free from defects and procured from acceptable Origin of Manufacturer by Project. The materials and equipment shall be accepted by PURCHASER and the Classification Society. Materials and equipment shall be manufactured according to internationally recognized standards for the offshore oil drilling and production industries, and shall be in conformance with the Basic Design and Agreement specifications and requirements.

All permanent materials and equipment received at the site must come with the proper material certificates. Materials of unknown origin or materials without any certification shall not be used unless agreed otherwise in writing by Company authorized representative.

Full material traceability with traceability sheet shall be provided by Supplier. For various systems consisting of numerous parts with possibly many different types and grades of materials, it is important that only the correct materials are connected and installed together in a system and equipment. For this to be achieved there must be auditable traceability of all components. There shall be a system in place that allows full traceability to an original material certificate with heat no. and Lot no. of every component part. The cast / heat number as marked, provides traceability back to the relevant material certification. Traceability sheet of material shall be kept to trace where the materials have been used along with installation location, basic description, material grade, heat number, certificate number, NDE record and welding records.

All permanent materials and equipment shall be traceable to the source via identification and marking inscribed or low pressure die stamped on the work piece. Supplier shall put in place a traceability procedure to ensure the proper traceability of the materials and equipment.

All materials purchased by and received by Supplier, must be tagged and checked

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against “Original Material certificates” or “Certified True copies’ of originals, on receipt.

On receipt of the materials and equipment, Supplier shall check against the shipping manifest and inspect the goods to ensure there is no missing and / or damaged items.

Supplier shall provide adequate and proper storage for the materials and equipment. Materials and equipment which are sensitive to environmental conditions shall be stored in covered storage areas and / or air-conditioned rooms. Space and anti-condensation heaters of equipment shall be energized while the equipment is in storage. All valves shall be stored in covered storage areas not exposed to blasting and painting activities prior to installation at site. Proper maintenance and greasing of the valves internal shall be provided during construction phase.

Fabrication of different materials such as carbon steel, alloy low temperature steel, corrosion resistant alloys (CRA), etc., shall be carried out in segregated areas to prevent cross contamination and mix-up.

Traceability Requirements for Materials / Disciplines is as below :

I. Definitions for traceability are as follows:

a) Full Traceability (“FT”) - For Materials classified as “FT” an exact correlation between every Material and its respective manufacturing certificate shall be assured, for the duration of the work of this Agreement including after assembly;

b) Limited Traceability (“LT”) - For Materials classified as “LT” an exact correlation between every Material or its manufacturing lot and its respective manufacturing and testing certificate shall be assured, by the delivery at the PURCHASER job Site, during storage and at assembly. It is not necessary to maintain it after completion of the work of this Agreement;

c) Recognized Origin (“RO”) - For Materials classified as “RO” it shall be assured that the Material was purchased from a qualified supplier.

II. Material Certificates are required for all materials being received. Receiving Inspection shall be performed including the verification of the manufacturing certificate, according to the applicable standard.

III. Supplier shall follow the following rules for the traceability of materials per discipline. Traceability requirements for items not listed shall be agreed between PETROBRAS and Supplier.

A. General Requirements – All disciplines

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MATERIALS / COMPONENTS	FT	LT	RO
Bolts, Nuts, Washers (Note 1)	-	-	X
Gaskets (Note 2)	-	-	X
Insulating and refractory consumables	-	-	X
Painting consumables (paint, varnish and solvents)	-	X	-
NDE consumables	-	-	X
Welding Consumables (Note 3)	X	-	X
Welded Joints (Note 4)	X	-	-
Bolted Joints (Note 5)	X	-	-
Compression Joints - gasketed and non gasketed (Note 5)	X	-	-
Valves (manual or actuated)	X	-	-
Safety Valves	X	-	-
Fire Safe Materials	X	-	-
Composite Materials	-	-	X

Note 1: All bolts and nuts shall be properly hard stamped (by the manufacturer) according to the applicable standard. This is intended to properly identify and avoid misuse of regular bolts and special purpose bolts (e.g. low temperature, NACE resistant, high yield strength).

Note 2: All gaskets shall be properly marked and color coded as required by the applicable standards.

Note 3: FT for welding consumables is required for the Procedure Qualification Records (PQR) and for welded joints in special application, such as submarine pipelines.

Note 4: Welded joints to all disciplines shall necessarily be traceable to the welder, to the Welding Procedure Specification (WPS), and to the non-destructive evaluations reports applicable.

Note 5: Bolted joints and compressions joints shall be traceable to the professionals who have performed its assembly and tightening.

B. Piping

MATERIALS	FT	LT	RO
Pressure retaining parts (pipes, forgings, castings)	X (Note 5)	-	-
Expansion Joints	X	-	-
Special Supports	X	-	-
Spring Hangers	X	-	-
Hoses	-	-	X
Quick release couplings	-	-	X

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Note 5: FT is required for these materials on the following cases:

a) Carbon steel lines for temperature below -29ºC design temperature;

b) Category M fluid service, as defined in ASME B31.3;

c) Alloy steel (low, medium, high alloy, including stainless steel grades), and all non ferritic materials (Cu-Ni, Inconel, Titanium, Hastelloy, and so on).

For the remaining materials ("p-number 1" above -29ºC design temperature) LT shall suffice.

C. Static Equipment (pressure vessels, heat exchangers, tanks, filters, and so on)

MATERIALS	FT	LT	RO
Pressure retaining parts	X		
Internals	X		

D. Rotating Equipment

ROTATING EQUIPMENT	FT	LT	OR
Compressors	X		
Turbines	X		
Pumps	X		

Note 8: Materials are considered purchased individually, independently of equipment purchase. For materials that are part of the equipment, the supplier shall guarantee the traceability of every material.

H. Safety

MATERIALS	FT	LT	RO
Fire extinguishers			X
Hydraulic systems			X
Electronic systems			X
Mechanical Equipment			X
Life buoys and jackets			X
Flash lights			X
Light and smokes signaling devices			X
Metallic and non metallic fabric			X
Hoses			X
Masks, Fibber glass boxes			X

I. Architecture

MATERIALS	FT	LT	RO
Classified partition panels		X (note 7)	

Note 7: The supplier shall guarantee similarity of the supplied material in relation to the tested prototype.

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J. Mooring system

MATERIALS	FT	LT	RO
Hook up/ installation chains	X		
Fairleads	X		
Chain-stoppers	X		
Hook-up/installation wire ropes	X		

K. Structural Steel Components (Note 9)

MATERIALS	FT	LT	RO
Plates	X		
Profiles	X		

Note 9: RO may be applied to tertiary structures (hand rails, stairs, and so on)

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